



ALBUKHARY INTERNATIONAL UNIVERSITY

CHAPTER 4

MATHEMATICAL INDUCTION

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"inspiring minds"

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Learning Outcomes

After going through this lesson, you should be able to:

1. Prove various mathematical statements and theorems by mathematical induction.
2. Differentiate between strong induction and simple induction.
3. Understand induction by well-ordering principle
4. Understand and prove for recursive definition and structural induction.



4.1

Principle of Mathematical Induction

1. Mathematical induction is a concept in mathematics that is used to prove various mathematical statements and theorems.
2. For example, how to prove that

$$2 + 4 + 6 + \cdots + 2n = n(n + 1) \text{ for all natural numbers?}$$

Usually, we check one by one:

$$n = 1, 1(1 + 1) = 2$$

$$n = 2, 2(2 + 1) = 6$$

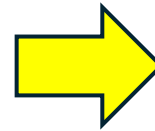
$$n = 3, 3(3 + 1) = 12$$

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Domino Effect



Formula True for the 1st Term



Formula True for the 2nd Term

⋮

What are the conditions that needs to be satisfied in playing dominos?

Condition 1

When the first domino falls, it will hit the second one

Condition 2

Each domino will hit the next one and each domino that is hit will fall

Similar to a mathematical induction:

Condition 1 (Base Case)

$P(1)$ is true for $n = 1$

Condition 2 (Inductive Case)

If $P(n)$ is true for $n = k$,
then $P(n)$ is true for $n = k + 1$

Example 1

For any natural number n , prove that $2 + 4 + 6 + \cdots + 2n = n(n + 1)$.

Solution:

Let $P(n) = 2 + 4 + 6 + \cdots + 2n = n(n + 1)$.

Condition 1: Base Case (Prove that $P(1)$ is true)

$$\text{LHS: } P(1) = 2$$

$$\text{RHS: } P(1) = 1(1 + 1) = 2$$

Hence, $P(1)$ is true.

Example 1 (cont.)

Solution:

Condition 2: Inductive Case (If $P(k)$ is true, then $P(k + 1)$ is also true)

$P(k)$

$$\text{LHS: } P(k) = 2 + 4 + 6 + \cdots + 2k$$

$$\text{RHS: } P(k) = k(k + 1)$$



$P(k + 1)$

$$\text{LHS: } P(k + 1) = 2 + 4 + 6 + \cdots + 2k + 2(k + 1)$$

$$= k(k + 1) + 2(k + 1)$$

$$= (k + 1)(k + 2)$$

$$\text{RHS: } P(k + 1) = (k + 1)((k + 1) + 1)$$

$$= (k + 1)(k + 2)$$

Now by PMI, we can say that $2 + 4 + 6 + \cdots + 2n = n(n + 1)$.

Example 2

For any natural number n , prove that $1 + 3 + 5 + \cdots + 2(n - 1) = n^2$.

Solution:

Let $P(n) = 1 + 3 + 5 + \cdots + 2(n - 1) = n^2$.

Condition 1: Base Case (Prove that $P(1)$ is true)

$$\text{LHS: } P(1) = 1$$

$$\text{RHS: } P(1) = 1^2 = 1$$

Hence, $P(1)$ is true.

Example 2 (cont.)

Solution:

Condition 2: Inductive Case (If $P(k)$ is true, then $P(k + 1)$ is also true)

$P(k)$

$$\text{LHS: } P(k) = 1 + 3 + 5 + \cdots + 2(k - 1)$$

$$\text{RHS: } P(k) = k^2$$



$P(k + 1)$

$$\text{LHS: } P(k + 1) = 1 + 3 + 5 + \cdots + 2(k - 1) + 2(k + 1) - 1$$

$$= k^2 + 2k + 1$$

$$\text{RHS: } P(k + 1) = (k + 1)^2$$

$$= k^2 + 2k + 1$$

Now by PMI, we can say that $1 + 3 + 5 + \cdots + 2(n - 1) = n^2$.

Example 3

For all $n \geq 1$, prove that $1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{(n(n+1)(2n+1))}{6}$.

Solution:

Condition 1: Base Case (Prove that $P(1)$ is true)

$$\text{LHS: } P(1) = 1^2 = 1$$

$$\text{RHS: } P(1) = \frac{(1(1+1)(2(1)+1))}{1} = 1$$

Example 3 (cont.)

Solution:

Condition 2: Inductive Case (If $P(k)$ is true, then $P(k + 1)$ is also true)

$P(k)$

$$\text{LHS: } P(k) = 1^2 + 2^2 + 3^2 + \dots + k^2$$

$$\text{RHS: } P(k) = \frac{(k(k+1)(2k+1))}{6}$$


$P(k + 1)$

$$\text{LHS: } P(k + 1) = 1^2 + 2^2 + 3^2 + \dots + k^2 + (k + 1)^2$$

$$= \frac{(k(k+1)(2k+1))}{6} + (k + 1)^2$$

$$= \frac{2k^3 + 9k^2 + 13k + 6}{6}$$

$$\text{RHS: } P(k + 1) = \frac{(k+1)(k+1+1)(2(k+1)+1))}{6}$$

$$= \frac{2k^3 + 9k^2 + 13k + 6}{6}$$

Now by PMI, we can say that $1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{(n(n+1)(2n+1))}{6}$.



4.2

Strong Induction and Well-Ordering Principle

Strong Induction

1. The main difference between ordinary (or weak) induction and strong induction lies in the inductive hypothesis, the assumption you make in the inductive step of the proof. In ordinary induction, you assume that the statement holds true for a specific value ' k ', and then you prove it holds for ' $k + 1$ '.
2. In **strong induction**, you assume that the statement holds true for all values from the base case up to ' k ', and then you prove it holds for ' $k + 1$ '

Condition 1 **(Base Case)**

$P(1)$ is true for $n = 1$

Condition 2 **(Inductive Case)**

The statement $P(n)$ is true for
from the base case up to k ,
then $P(n)$ is true for $n = k + 1$

If we think about this from the domino analogy, with Weak Induction we show that one domino knocks down the next. With strong induction we show that if all the previous dominoes have been knocked over, the next one will also fall.

Example 4

Given $n \in \mathbb{N}$, define a_n recursively as follows:

$$a_0 = 1, a_1 = 3, a_n = 2a_{n-1} - a_{n-2} \text{ for } n \geq 2.$$

Prove that for all $n \geq 0$, $a_n = 2n + 1$.

Proof. We work by strong induction on n .

For the base cases, first consider $n = 0$. We have that $a_0 = 1 = 2(0) + 1$, so the result is true when $n = 0$.

Next, consider $n = 1$. We have that $a_1 = 3 = 2(1) + 1$, so the result is true when $n = 1$.

For the strong inductive hypothesis, suppose that for some $n \geq 1$, we have that $a_k = 2k + 1$ for all $0 \leq k \leq n$. Consider a_{n+1} . We have

$$\begin{aligned} a_{n+1} &= 2a_n - a_{n-1} && \text{(because } n+1 \geq 2) \\ &= 2(2n+1) - (2(n-1)+1) && \text{(by the strong inductive hypothesis)} \\ &= 4n+2 - 2n+2 - 1 \\ &= 2n+3 \\ &= 2(n+1)+1. \end{aligned}$$

Therefore, the result holds for $n+1$ as well.

Hence, by the Principle of Strong Induction, we have $a_n = 2n + 1$ for all $n \geq 0$. □

When to apply simple induction or strong induction?

Simple Induction:

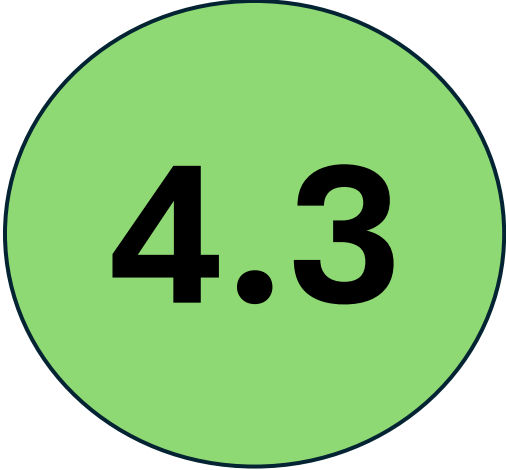
Use when a statement at $n + 1$ logically depends only on the validity at n .

Strong Induction:

Use when the step from n to $n + 1$ involves dependencies on multiple preceding cases. For instance, in recursive algorithms that combine earlier computed elements (e.g., $F(n) = F(n - 1) + F(n - 2)$).

Well-Ordering Principle

1. The **Well-Ordering Principle** states that every nonempty set of non-negative integers contains a least element.
2. In simpler terms, it means that if you take any collection of whole numbers (including zero and positive integers), there will always be a smallest number within that collection.
3. This principle is foundational in mathematics, particularly in relation to mathematical induction and the construction of integers.



4.3

Recursive Definition and Structural Induction

Recursive Definition

1. Sometimes it is difficult to define an object explicitly. However, it may be easy to define this object in terms of itself. This process is called recursion.
2. Definition of recursive data types have two parts:

Step 1 **(Base Case)**

Specify the value of the function at zero.

Part 2 **(Constructor Case)**

Specifying how to construct new data elements from previously constructed elements or from base elements

Example 5

Calculate the values of this function for $f(0)$, $f(1)$, $f(2)$ and $f(3)$.

$$f(0) = 5$$

$$f(n) = f(n - 1) + 2$$

Solution:

$$f(0) = 5$$

$$f(1) = f(1 - 1) + 2 = 7$$

$$f(2) = f(2 - 1) + 2 = 9$$

$$f(3) = f(3 - 1) + 2 = 11$$

Example 6

Calculate the values of this function for $f(0)$, $f(1)$, $f(2)$ and $f(3)$.

$$f(0) = 0$$

$$f(n) = f(n - 1) + 2n - 1$$

Solution:

$$f(0) = 0$$

$$f(1) = f(1 - 1) + 2(1) - 1 = 1$$

$$f(2) = f(2 - 1) + 2(2) - 1 = 4$$

$$f(3) = f(3 - 1) + 2(3) - 1 = 9$$

Structural Induction

Structural induction is a method for proving that all the elements of a recursively defined data type have some property.

A structural induction proof has two parts corresponding to the recursive definition:

Part 1 **(Base Case)**

Prove that each base case element has the property

Part 2 **(Constructor Case)**

Prove that each constructor case element has the property, when the constructor is applied to elements that have the property.

Example 7

Suppose a set S is defined recursively as follows:

- (a) $1 \in S$
- (b) If $x \in S$ then $2x \in S$

Prove, using **mathematical induction**, that every element of S is a power of 2.

Solution:

Starting from 1, you can only keep doubling:

$$1 \in S \text{ or } 2 \in S \text{ or } 4 \in S \text{ or } 8 \in S$$

So, the set is exactly

$$S = \{2^n | n \in \mathbb{N}\} = \{1, 2, 4, 8, 16, \dots\}$$

(a) Base Case:

Observe that $1 = 2^0$ so 1 is an integer power of 2

(b) Inductive Case:

Suppose $x \in S$ and x is an integer power of 2. We claim that $2x$ is an integer power of 2 as well. Since x is a power of 2, we know $x = 2^k$ for some $k \in \mathbb{Z}$. Then, $2x = 2(2^k) = 2^{k+1}$ and since $k + 1 \in \mathbb{Z}$ we know that 2^x is also a power of 2.

Example 8

Suppose a set S is defined recursively as follows:

- (a) $(0,0) \in S$
- (b) If $(x, y) \in S$ then $(x, y + 1), (x + 1, y + 1), (x + 2, y + 1) \in S$

Let's prove that every element $(x, y) \in S$ has $x \leq 2y$.

Solution:

(a) Base Case:

Observe that $(0,0)$ certainly satisfies $0 \leq 2(0)$.

(b) Inductive Case:

1. We need to prove that $(x, y + 1)$ satisfies $x \leq 2(y + 1)$. However since $x \leq 2y \leq 2y + 2$, this is true.
1. We need to prove that $(x + 1, y + 1)$ satisfies $x + 1 \leq 2(y + 1)$. However since $x + 1 \leq 2y + 1 \leq 2y + 2$, this is true.
2. We need to prove that $(x + 2, y + 1)$ satisfies $x + 2 \leq 2(y + 1)$. However since $x + 2 \leq 2y + 2$, this is true.



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Thank You

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